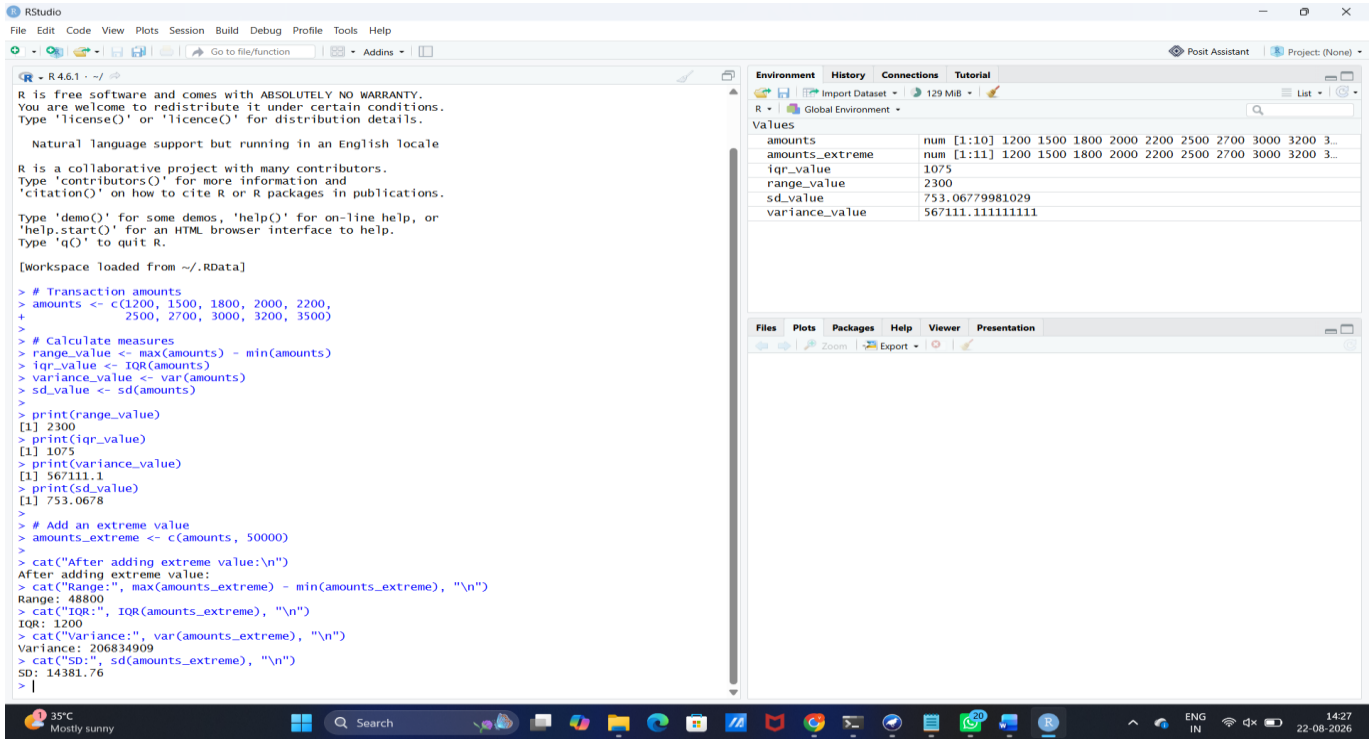


SET 15

1. R: Calculate range, IQR, variance and SD for 10 transaction amounts and discuss sensitivity to an extreme value.



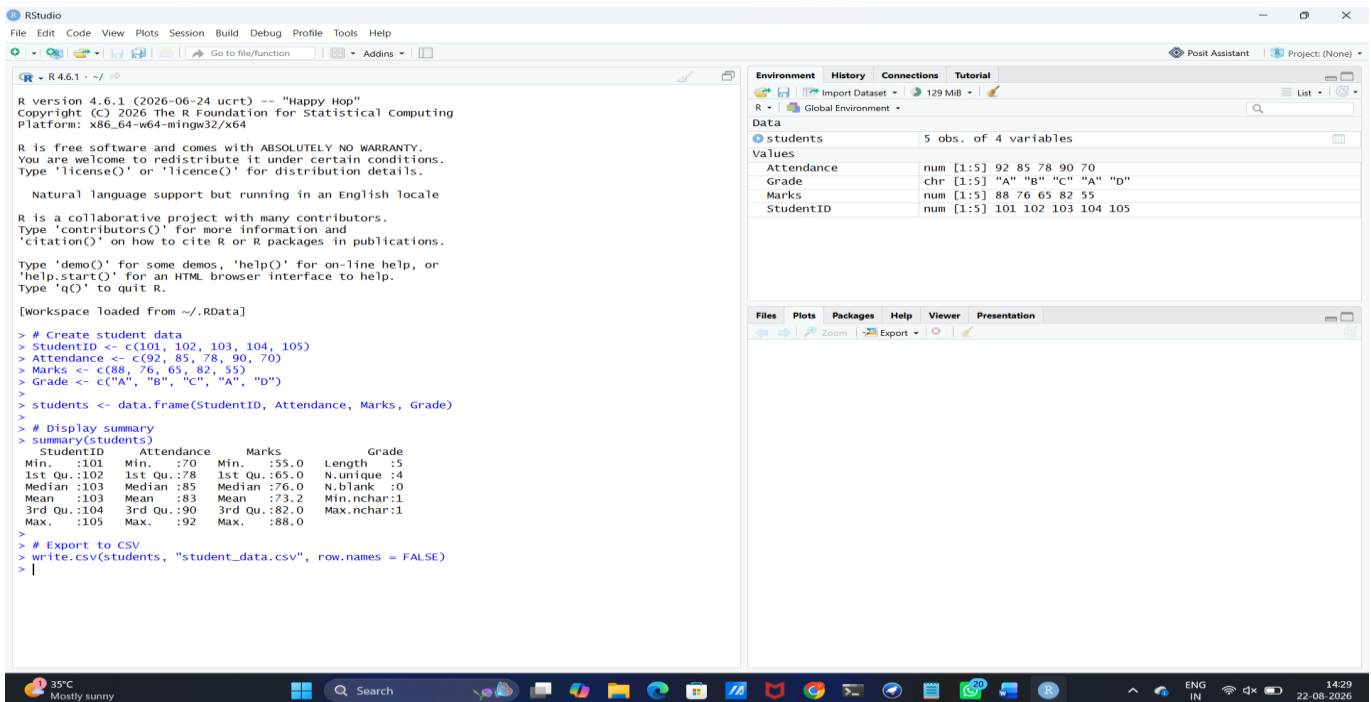
The screenshot shows the RStudio interface. The console on the left displays the following R code and its output:

```
R - R 4.6.1 ~ /  
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
[Workspace loaded from ~/.RData]  
> # Transaction amounts  
> amounts <- c(1200, 1500, 1800, 2000, 2200,  
+ 2500, 2700, 3000, 3200, 3500)  
>  
> # Calculate measures  
> range_value <- max(amounts) - min(amounts)  
> iqr_value <- IQR(amounts)  
> variance_value <- var(amounts)  
> sd_value <- sd(amounts)  
>  
> print(range_value)  
[1] 2300  
> print(iqr_value)  
[1] 1075  
> print(variance_value)  
[1] 567111.1  
> print(sd_value)  
[1] 753.0678  
>  
> # Add an extreme value  
> amounts_extreme <- c(amounts, 50000)  
>  
> cat("After adding extreme value:\n")  
After adding extreme value:  
> cat("Range:", max(amounts_extreme) - min(amounts_extreme), "\n")  
Range: 48800  
> cat("IQR:", IQR(amounts_extreme), "\n")  
IQR: 1200  
> cat("Variance:", var(amounts_extreme), "\n")  
Variance: 206834909  
> cat("SD:", sd(amounts_extreme), "\n")  
SD: 14381.76  
> |
```

The Environment pane on the right shows the following values:

Variable	Value
amounts	num [1:10] 1200 1500 1800 2000 2200 2500 2700 3000 3200 3500
amounts_extreme	num [1:11] 1200 1500 1800 2000 2200 2500 2700 3000 3200 3500 50000
iqr_value	1075
range_value	2300
sd_value	753.06779981029
variance_value	567111.111111111

2. R: Create StudentID, Attendance, Marks and Grade data; use summary() and export CSV.



The screenshot shows the RStudio interface. The console on the left displays the following R code and its output:

```
R - R 4.6.1 ~ /  
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"  
Copyright (C) 2026 The R Foundation for Statistical Computing  
Platform: x86_64-w64-mingw32/x64  
  
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
[Workspace loaded from ~/.RData]  
> # Create student data  
> StudentID <- c(101, 102, 103, 104, 105)  
> Attendance <- c(92, 85, 78, 90, 70)  
> Marks <- c(88, 76, 65, 82, 55)  
> Grade <- c("A", "B", "C", "A", "D")  
>  
> students <- data.frame(StudentID, Attendance, Marks, Grade)  
>  
> # Display summary  
> summary(students)  
StudentID Attendance Marks Grade  
Min. :101 Min. :70 Min. :55.0 Length :5  
1st Qu.:102 1st Qu.:78 1st Qu.:65.0 N.unique :4  
Median :103 Median :85 Median :76.0 N.blank :0  
Mean :103 Mean :83 Mean :73.2 Min.nchar:1  
3rd Qu.:104 3rd Qu.:90 3rd Qu.:82.0 Max.nchar:1  
Max. :105 Max. :92 Max. :88.0  
>  
> # Export to CSV  
> write.csv(students, "student_data.csv", row.names = FALSE)  
> |
```

The Environment pane on the right shows the following data:

Variable	Value
students	5 obs. of 4 variables
Attendance	num [1:5] 92 85 78 90 70
Grade	chr [1:5] "A" "B" "C" "A" "D"
Marks	num [1:5] 88 76 65 82 55
StudentID	num [1:5] 101 102 103 104 105

3. WEKA: Create telecom ARFF data (Age, MonthlyCharge, ContractType, Churn); apply J48 and report confusion matrix.

Dataset: Telecom Churn

```
@relation telecom_churn
@attribute Age numeric
@attribute MonthlyCharge numeric
@attribute ContractType {Monthly,Yearly}
@attribute Churn {Yes,No}
@data
22,45,Monthly,Yes
28,55,Monthly,Yes
35,60,Yearly,No
42,70,Yearly,No
25,50,Monthly,Yes
38,65,Yearly,No
31,58,Monthly,No
47,75,Yearly,No
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Cross-validation' with 8 folds. The result list shows '14:31:38 - treesJ48' as the selected result. The classifier output displays the following statistics:

Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	6	75 %
Incorrectly Classified Instances	2	25 %
Kappa statistic	0.4667	
Mean absolute error	0.25	
Root mean squared error	0.5	
Relative absolute error	47.3684 %	
Root relative squared error	92.8279 %	
Total Number of Instances	8	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.667	0.200	0.667	0.667	0.667	0.467	0.733	0.569	Yes
	0.800	0.333	0.800	0.800	0.800	0.467	0.733	0.765	No
Weighted Avg.	0.750	0.283	0.750	0.750	0.750	0.467	0.733	0.692	

=== Confusion Matrix ===

```
a b <-- classified as
2 1 | a = Yes
1 4 | b = No
```

4. WEKA: Apply Naive Bayes to telecom data and compare correct percentage and Kappa with J48.

Dataset: Telecom Churn – Naïve Bayes

Use the same Telecom Churn ARFF dataset from Set 15 Q3.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseNaiveBayes

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds8

☐ Percentage split

%66

More options...

(Nom) Churn

StartStop

Result list (right-click for options)

14:31:38 - trees.J48

14:32:27 - bayes.NaiveBayes

Classifier output

Monthly4.02.0

Yearly1.05.0

[total]5.07.0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances675%

Incorrectly Classified Instances225%

Kappa statistic0.4667

Mean absolute error0.2254

Root mean squared error0.4408

Relative absolute error42.7054%

Root relative squared error81.8347%

Total Number of Instances8

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.667	0.200	0.667	0.667	0.667	0.467	0.933	0.917	Yes
	0.800	0.333	0.800	0.800	0.800	0.467	0.933	0.967	No
Weighted Avg.	0.750	0.283	0.750	0.750	0.750	0.467	0.933	0.948	

=== Confusion Matrix ===

a b <-- classified as

2 1 | a = Yes

1 4 | b = No

StatusOK

Log

x 0